

Faculty of Engineering Technology SPU	Al-Balqa Applied University 	Department of Scientific Basic Sciences April 18, 2022
مدرس المادة: <u>د. كمال فاضل</u>	موعد المحاضرة: <u>8-9</u> <u>20.2</u>	اسم الطالب: <u>أحمد أيوب</u>

Numerical Techniques and Applied Numerical Analysis -Midterm Exam. (30 Marks)
Allowed Time: 50 minutes

Question	1	2	3	4	5	6	7	8	9	10
Answer	a	b	c	c	d	b	c	b	a	b

Part 1- Choose the correct answer (20 points)

- If the estimated length of a bridge equals 19m, if its true length is 20m, then the percent relative true is:
 (a) 5% b. 6% c. 10% d. 8%
- Using the Newton-Raphson method; approximate the root of the equation; $x^2 = e^{-x}$, by rounding off (x_3) to 4 decimal places. Given that the initial guess; $x_0 = 0.5$?
 a. 0.7034 (b) 0.7035 c. 0.7036 d. 0.7037
- Given that (1) is a root of the equation; $x^5 - 2x^4 + 4x^3 - 6x^2 + 3x = 0$, then its multiplicity is equal to:
 (a) 1 b. 3 c. 2 d. 4
- By the simple fixed-point method, approximate the root of the equation, $x + \ln(x + 1) = 5$, after rearranging it as $x = 5 - \ln(x + 1)$, $x_0 = 1$.
 Do two iterations and round off your answer to 4 decimal places.
 a. 4.3068 b. 4.3069 (c) 3.3310 d. 3.3311
- Normalize the number (0.000 000 0568) "write it in the floating decimal form"?
 a. 0.568×10^7 b. 5.68×10^6 c. 5.68×10^{-6} (d) 0.568×10^{-7}
- If the true value of Y is (15) and its approximations were:
 7, 7.1, 6.9, 7.2, 6.8, 7.3 this means that we have
 a. high accuracy and low precision (b) low accuracy and high precision
 c. high accuracy and high precision d. low accuracy and low precision

7. If n is a positive integer such that the equation; $2x^3 - 5x^2 + 4x - 10 = 0$, has a root in the interval $[n, n + 1]$, then $n =$

- a. 1 b. 4 c. 2 d. 3

8. The second non-zero term of the Maclaurin series corresponding to the function;

$f(x) = 3x^2 \sin 2x$, is equal to:

- a. $-4x^3$ b. $4x^3$ c. $4x^5$ d. $-4x^5$

9. Which of the following is a bracketing method?

- a. Bisection method b. Newton-Raphson Method
c. Simple fixed-point method d. Secant method

10. If the curve of the polynomial f touches the x-axis at $x = a$ then

- a. $f'(a) > f(a)$ b. $f'(a) < f(a)$
c. $f'(a) = f(a)$ d. $f'(a)f(a) < 0$

Part Two: Essay Questions (10 Points)

(حل هذه الاسئلة على الصفحات الفارغة ولا تكتب عليها شيء آخر)

1. The volume of a cube is 10 cm^3 . Use bisection method twice to approximate its side, take the interval $[2,3]$? (3 points)

2. Use the first four non-zero terms of Maclaurin expansion corresponding to e^x , to approximate the value of e^3 ? (3 points)

3. Given that $f(x) = \frac{3}{4}x^4 - \frac{1}{3}x^3 + \frac{3}{2}x^2 - x + 1$.

By Bolzano theorem, show that $f'(x) = 0$

has a root in the interval $(0,1)$, then approximate it by false position method by doing one iteration? (4 points)

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1- ~~the~~

V : Volume

L : side

$$V = L^3$$

$$\Rightarrow f(L) = V - L^3 = 10 - L^3$$

$\Rightarrow$ the interval $[2, 3]$

$$\textcircled{1} x_L = 2 \Rightarrow f(x_L) = 2$$

$$x_u = 3 \Rightarrow f(x_u) = -17$$

$\Rightarrow$ the function continuous

$$\textcircled{2} f(x_u) f(x_L) < 0$$

Bolzano $\checkmark$

$$\Rightarrow x_r = \frac{2+3}{2} = 2.5$$

$$f(2.5) = -5.625$$

$$\textcircled{2} x_L = 2 \Rightarrow f(x_L) = 2$$

$$x_u = 2.5 \Rightarrow f(x_u) = -5.625$$

$f(x_u) f(x_L) < 0$ Bolzano $\checkmark$

$$x_r = \frac{2 + f(2.5)}{2} = \boxed{2.25}$$

$$f(x_r) = -1.390625$$

2- ~~the~~

$$f(x) = e^x = f(3) = e^3$$

$$f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3$$

$$f(x) = e^x \Rightarrow f(0) = 1$$

$$f(x) = e^x \Rightarrow f'(0) = 1$$

$$f(x) = e^x \Rightarrow f''(0) = 1$$

$$f(x) = e^x \Rightarrow f'''(0) = 1$$

$$\Rightarrow f(3) = 1 + 3 + \frac{(3)^2}{2} + \frac{(3)^3}{6}$$

$$\boxed{f(3) = e^3 = 13}$$

3

السؤال (3)

$[0, 1]$

$$f(x) = \frac{3}{4}x^4 - \frac{1}{3}x^3 + \frac{3}{2}x^2 - x + 1$$

$$f'(x) = 3x^3 - x^2 + 3x - 1$$

$$\Rightarrow f(0) = -1$$

$$f'(1) = 3 - 1 + 3 - 1 = 4$$

$$(1) f(0) f'(1) < 0$$

(2) The function is continuous

$\Rightarrow$ Bolzano ✓

$$x_1 = 0 \Rightarrow f(x_1) = -1$$

$$x_n = 1 \Rightarrow f(x_n) = 4$$

$$x_r = x_n - \frac{f(x_n) - f(x_1)}{f'(x_n) - f'(x_1)}$$

$$x_r = 1 - \frac{4 - (-1)}{4 - (-1)}$$

$$x_r = 1 - \frac{4(0 - 1)}{(-1 - 4)}$$

$$x_r = 0.2$$

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